



جامعة قناة السويس
كلية الهندسة



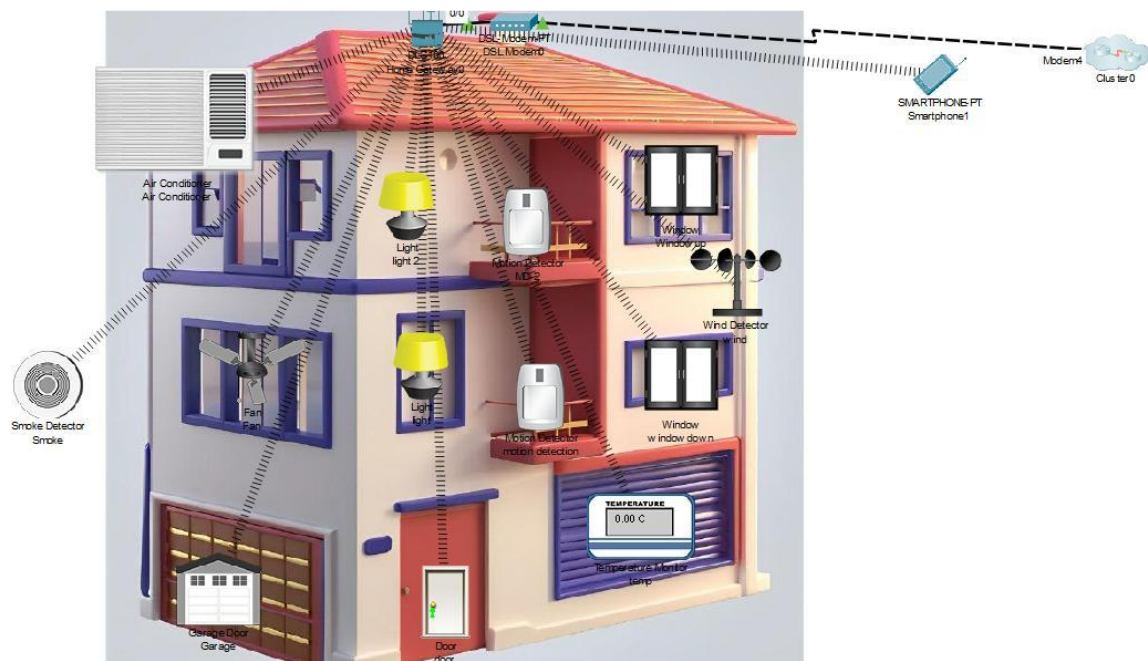
**Project:
SMART-HOME**

**Course Name:
Wireless Network**

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Smart Home Simulation Project Report Using Cisco Packet Tracer



1. Introduction

This project presents a simulation of a smart home environment using Cisco Packet Tracer. The smart home is designed to mimic a real-world IoT-enabled residence consisting of two floors, a garage, and a roof. The primary goal of this simulation is to demonstrate the integration and automation of various smart devices, controlled remotely through a smartphone connected to the internet via a cloud infrastructure.

2. Project Overview

2.1. Layout and Components

The smart home is structured across multiple zones:

• Garage:

- Automatic Gate
- Fire Alarm

• First Floor:

- Smart Light
- Smart Fan
- Temperature Sensor
- Smart Window
- Motion Detector

• Second Floor:

- Smart Light
- Air Conditioner (AC)
- Smart Window
- Motion Detector

- **Roof:**

- Wind Detector

- **Networking and Control:**

- Home Gateway
- Smartphone (used as remote controller)
- Cloud (Internet connectivity)
- ISP Server
- DNS Server
 - Google Server (simulating internet access)



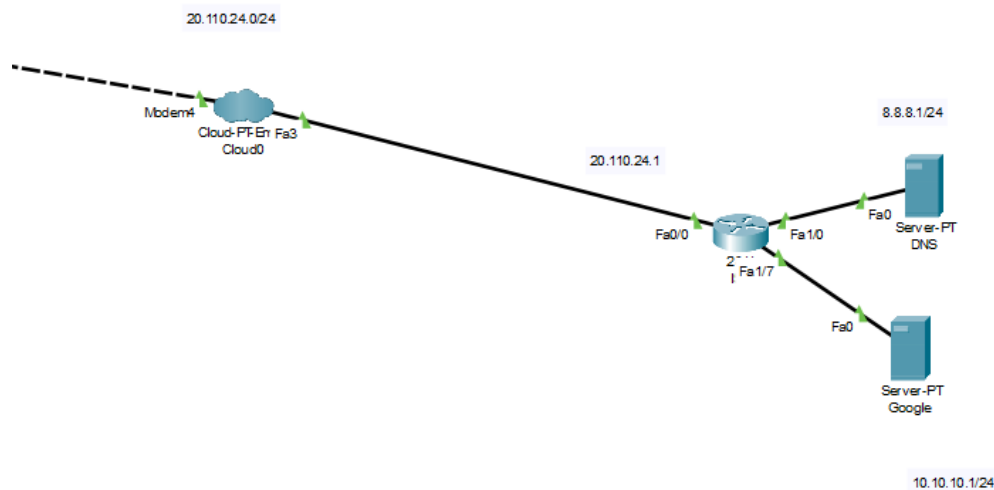
The screenshot shows a web browser window titled "IoT Monitor" with a close button (X) in the top right corner. Below the title bar, the page header reads "IoT Server - Devices" on the left and navigation links "Home | Conditions | Editor | Log Out" on the right. The main content area is a table listing 14 IoT devices, each with a green status indicator, a device name with ID, and a function name.

Device Name (ID)	Function
Garage (PTT08102PQ6-)	Garage Door
light (PTT0810W3VJ-)	Light
motion detection (PTT08108WXN-)	Motion Detector
Fan (PTT0810R69J-)	Ceiling Fan
Air Conditioner (PTT08106Q4O-)	AC
door (PTT081023JN-)	Door
Windowup (PTT0810AU6G-)	Window
windowdown (PTT0810H3Y2-)	Window
MD-2 (PTT08106T2G-)	Motion Detector
light 2 (PTT08102UNY-)	Light
Smoke (PTT08102639-)	Smoke Detector
temp (PTT08103Z66-)	Temperature Monitor
wind (PTT0810YVH0-)	Wind Detector

3. Network Architecture

The smart home devices are connected via the **Home Gateway**, which communicates with the **Cloud** to provide internet connectivity. The **ISP Server** connects the cloud to external services such as **Google** and a **DNS Server** to resolve domain names.

The **Smartphone** is configured to remotely monitor and control smart devices using the internet connection, simulating a real-time user interface.



4. Automation Logic

Two core automation conditions are implemented in this simulation using logical programming and device interaction:

4.1. Motion-Based Automation

- **Trigger:** Motion detected by the motion sensors on the 1st and 2nd floors.

- **Actions:**

- On the **1st floor**, motion triggers:

- Light ON

- Fan ON

- On the **2nd floor**, motion triggers:

- Light ON

- Air Conditioner ON

4.2. AC-Window Dependency

- **Condition:** When the **Air Conditioner** is turned ON (automatically or manually).

- **Action:** The **Windows on the 2nd floor** automatically **close** to preserve the internal temperature.

IoT Monitor						X
IoT Server - Device Conditions				Home Conditions Editor Log Out		
Actions		Enabled	Name	Condition	Actions	
Edit	Remove	Yes	Enter_floor1	motion detection On is true	Set light Status to On Set Fan Status to High	
Edit	Remove	Yes	Enter_floor2	MD-2 On is true	Set light 2 Status to On Set Air Conditioner On to true	
Edit	Remove	Yes	aircond_f2	Air Conditioner On is true	Set Windowup On to false	

5.

Device Communication and Control

The smartphone acts as a central controller, capable of manually overriding automated functions. All smart devices are configured with appropriate IP addresses and communication protocols .The DNS server resolves domain names, and access to services like Google ensures simulation of external internet interaction.

Web Browser

X

<

>

URL

http://192.168.25.1

Go

Stop

Home Gateway Login

Username:

Password:

Submit

Web Browser

X

<

>

URL

http://www.google.com

Go

Stop

Cisco Packet Tracer

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:

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6. Simulation Results

Upon running the simulation in Packet Tracer, the following outcomes were observed:

- Motion detection reliably triggered corresponding appliances.
- Temperature regulation and environmental control through window and AC coordination functioned as expected.
- The smartphone interface successfully interacted with devices remotely.
 - All components communicated efficiently through the internet infrastructure.